# TWISTED CONJUGACY IN IRREDUCIBLE SPHERICAL ARTIN GROUPS

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Abstract. We prove that the Artin groups of types $D_5$, $E_6$, $E_7$, $E_8$, $F_4$, $H_3$ and $H_4$ have property $R_\infty$. Together with previously known cases, this shows that every irreducible Artin group of spherical type and rank $\geq 2$ has $R_\infty$.

The proof uses the characteristic pure subgroup and its abelianization, viewed as the permutation lattice on the reflections. Rational, absolutely irreducible constituents of multiplicity one yield a fixed vector after composing with a suitable inner automorphism. The finite action that must be controlled is governed by automorphisms of Coxeter groups, so no description of $\mathrm{Aut}(A)$ is required. The constituents are explicit: eigenspaces of strongly regular graphs for $E_n$, frame modules for $F_4$ and $H_3$, and matrix coefficients in the icosian model for $H_4$.

The same argument gives uniform proofs in types $A_n$, $B_n$ and $D_n$; for $D_5$ this case appears to be new.

**[Contribution statement — placeholder, to be finalised by the authors.]** This work was developed with substantial assistance from AI systems (Anthropic's Claude) for the mathematical argument, the programs, the Lean formalisation and the text. So far it has been reviewed only by AI systems: an internal audit, two rounds of an AI "cold referee", and an external AI review of the manuscript. None of this is journal peer review, and no human expert has yet checked the work. The authors must replace this paragraph with an accurate statement of contributions before submission or posting.

## 1. Introduction

Let $G$ be a group and $\varphi \in \mathrm{Aut}(G)$. Elements $x, y \in G$ are *$\varphi$-twisted conjugate* if $y = zx\varphi(z)^{-1}$ for some $z \in G$. The number $R(\varphi) \in \{1, 2, \ldots, \infty\}$ of classes is the *Reidemeister number*, and $G$ has *property* $R_\infty$ if $R(\varphi) = \infty$ for every $\varphi \in \mathrm{Aut}(G)$ [17, 15].

For an Artin group $A = A(\Gamma)$ with Coxeter group $W = W(\Gamma)$, the question "which Artin groups have $R_\infty$?" is Problem 8.6 of the AIM problem list on Artin groups [1]. Known cases include:

- braid groups [14] and pure braid groups [9];
- the spherical types $A_n$, $B_n$, $D_4$, $I_2(m)$ [5];
- $D_n$ for $n \geq 6$ [30, Thm. 1];
- many non-spherical types [24, 30, 29].

The case $D_5$ was left open in [30]. The authors write there: "The only remaining case in the family $A[D_n]$ is $A[D_5]$. At present we do not have a description of $\mathrm{Aut}(A[D_5])$ ...; we conjecture though that both $A[D_5]$ and $A[D_5]/Z(A[D_5])$ have property $R_\infty$" [30, §1]. Paris–Soroko later computed $\mathrm{Aut}(A[D_5])$ [28], but they do not discuss $R_\infty$. (The survey list in [29, §1] attributes $D_n$, $n \geq 5$, to [30]; as far as we can see, $D_5$ is not proved there.) For spherical types the known proofs use an explicit description of $\mathrm{Aut}(A)$ or $\mathrm{Aut}(A/Z(A))$. For $I_2(m)$ they use instead the virtual freeness of $A/Z(A)$ [5].

**Main Theorem.** *The Artin groups of types $D_5$, $E_6$, $E_7$, $E_8$, $F_4$, $H_3$ and $H_4$ have property $R_\infty$.*

For $E_6, \ldots, H_4$ see Section 4. For $D_5$ see Proposition 5.2 and Remark 5.3, where we also note a short alternative derivation from [28, Thm. 1].

**Corollary.** *Every irreducible Artin group of spherical type and rank $\geq 2$ has property $R_\infty$.*

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The corollary follows from the Main Theorem together with [5] (types $A_n, B_n, D_4, I_2(m)$) and [30, Thm. 1] ($D_n$, $n \geq 6$). Rank one must be excluded: $A(A_1) \cong \mathbb{Z}$, and inversion on $\mathbb{Z}$ has Reidemeister number 2. The method also gives uniform proofs for $A_n$, $B_n$ and $D_n$ (Section 5), so among the irreducible spherical types only $I_2(m)$ needs a different argument (Remark 5.5).

*Significance and novelty.* For the six exceptional types $\mathrm{Aut}(A)$ is not known (cf. [28, §1]). The general criteria based on hyperbolic actions of $\mathrm{Aut}(A)$, or on Aut-invariant quasimorphisms [18, 19, 31], also require control of $\mathrm{Aut}(A)$.

The Main Theorem would follow from Conjecture (NA) of [30, §6]: finitely generated, residually finite, non-amenable groups have $R_\infty$. This appears as a theorem in [16], but its proof is reported to be flawed [30, §6]. Our results verify (NA), and Conjecture (R) [25, Problem 19.28], for these groups.

The broad strategy of obtaining an eigenvalue one via representation theory has precedents:

- Dekimpe–Gonçalves–Ocampo [9], for pure braid groups;
- Dekimpe–De Rock–Penninckx [8, Thm. 4.7], for flat manifolds. They require a $\mathbb{Q}$-irreducible $\mathbb{Z}$-subrepresentation $\eta_0$ of the holonomy representation whose image is not $\mathbb{Q}$-conjugate to that of any other $\mathbb{Q}$-subrepresentation (in particular, it has multiplicity one), and such that for every $D_0$ in the integral normaliser of $\eta_0(F)$ some $\eta_0(A)D_0$, $A \in F$, has eigenvalue one;
- the determinant criterion of Dekimpe–Penninckx for crystallographic groups [11, Cor. 3.10], as stated in [10, Thm. 3.3];
- the eigenvalue-one property of finite groups announced by Hiss–Lutowski [23]. For odd-dimensional constituents this gives an alternative route to the witnesses.

What is new here is:

- the application to Artin groups through the *characteristic reflection lattice* $H_1(P;\mathbb{Z}) \cong \mathbb{Z}^{\mathcal{T}}$;
- the control of the relevant finite symmetries by $\mathrm{Aut}(W)$ [21];
- the explicit sign and swap witnesses;
- the completion of the exceptional cases, and of $D_5$.

*Overview.* Let $P = \ker(A \to W)$, let $\mathcal{T}$ be the set of reflections, and let $\rho\colon W \to \mathrm{Sym}(\mathcal{T})$ be the conjugation action.

- $P$ is characteristic, and $H_1(P;\mathbb{Z}) \cong \mathbb{Z}^{\mathcal{T}}$ is a permutation module (Section 2).
- An automorphism $\varphi$ acts on $\mathbb{Z}^{\mathcal{T}}$ as $\pi N$, where $\pi$ realises the induced automorphism of $\rho(W) \cong W/Z(W)$ and $N$ commutes with $\rho(W)$. On a good constituent $U$, $N = \pm 1$.
- For a suitable inner automorphism $\iota_a$, the composite $\varphi' = \iota_a \circ \varphi$ has a fixed vector. So $R(\varphi'|_P) = \infty$, hence $R(\varphi') = \infty$, and $R(\varphi) = R(\varphi')$ (Section 3).
- The possible $\pi$ are controlled by $\mathrm{Aut}(W)$. Explicit constituents and witnesses finish the proof (Section 4).

*Formal verification.* A Lean 4/Mathlib formalisation accompanies this paper. It contains no `sorry`, no `axiom` and no `native_decide`. The Lean kernel checks:

- the Reidemeister lemmas and the criterion (both cases);
- the Artin group as a presented group, the map $f$, and the equivariant surjection $P^{ab} \to \mathbb{Z}^{\mathcal{T}}$;
- for each exceptional type and for $D_5$, finite certificates (commutant-scalar constituents and $\pm 1$ witnesses);
- the finite part of Lemma 4.1: by a search-tree (orbit–stabiliser) certificate, every reflection-preserving automorphism of $\rho(W)$ lies in $\rho(W)\langle\pi_0\rangle$, for all seven types ($\pi_0 = 1$ for $D_5$, $E_6$, $E_7$, $E_8$).

For each of the seven types the final theorem has exactly four hypotheses, the cited inputs: $W$ is finite; $P$ is characteristic [7]; $P^{ab} \cong \mathbb{Z}^N$ as groups, with $N = |\mathcal{T}|$ (Brieskorn); and the Franzsen–Howlett decomposition $\mathrm{Aut}(W) = R(W)\mathcal{A}(W)$ [21, Thm. 31]. The formalisation uses computer-found constituents rather than the hand constructions of Section 4. Type $D_5$ is formalised with

the same four hypotheses (for $D_5$ the Franzsen–Howlett input is [21, Thm. 31(3)]); its certificate uses the constituent $U^-$ of Proposition 5.2.

## 2. The characteristic reflection lattice

**Lemma 2.1.** *Let $\psi \in \mathrm{Aut}(G)$.*

*(a) If $K \trianglelefteq G$ with $\psi(K) = K$, then $R(\psi) \geq R(\bar{\psi})$.*
*(b) If $G = \mathbb{Z}^m$ and $\psi$ is given by $B$, then $R(\psi) = |\operatorname{coker}(I - B)|$. This is $\infty$ if $B$ has eigenvalue 1.*
*(c) $R(\iota_g \circ \psi) = R(\psi)$, where $\iota_g(x) = gxg^{-1}$.*
*(d) If $[G : H] < \infty$ and $\psi(H) = H$, then $R(\psi|_H) \leq [G : H]\, R(\psi)$.*

*Proof.* (a) The projection maps classes onto classes.
(b) $y \sim x$ iff $y - x \in (I - B)\mathbb{Z}^m$.
(c) $x \mapsto xg$ maps $\iota_g\psi$-classes bijectively to $\psi$-classes.
(d) Write $G = \bigsqcup_i Hg_i$. If $y = zx\psi(z)^{-1} \in H$ with $z = hg_i$, then $y$ is $\psi|_H$-conjugate to $g_i x \psi(g_i)^{-1}$. □

*Notation.* For a Coxeter graph with vertex set $S$:

- $A = \langle S \mid \Pi(s,t;m_{st}) = \Pi(t,s;m_{st})\rangle$, where $\Pi(s,t;m) = sts\cdots$ has $m$ letters;
- $W = A/\langle\langle s^2 \rangle\rangle$, with quotient map $\mu\colon a \mapsto \bar{a}$, and $P = \ker \mu$;
- $\mathcal{T}$ is the set of reflections, $\rho(w)(r) = wrw^{-1}$, and $\rho(w)$ also denotes the permutation matrix on $\mathbb{Z}^{\mathcal{T}}$.

Since $\mathcal{T}$ generates $W$, $\ker \rho = Z(W)$. Each $\alpha \in \mathrm{Aut}(W)$ induces $\theta_\alpha$ on $\rho(W)$ by $\theta_\alpha\rho(w) = \rho(\alpha w)$. For finite irreducible $W$ of rank $\geq 3$, $Z(W) \subseteq \{\pm 1\}$, and $r \mapsto \rho(r)$ is injective on $\mathcal{T}$.

**Theorem 2.2** ([7, Thm. 1.4])**.** *If $A$ is irreducible of spherical type, then $P$ is characteristic in $A$.*

This was extended to reducible types in [20]. The proof in [7] is computer-assisted in several cases:

- for $H_4, F_4, E_6, E_7, E_8$ it uses GAP calculations [7, Prop. 5.4, Cor. 5.5];
- for $H_3$ it uses one GAP calculation [7, Lemma 5.1];
- among the classical types it uses GAP for $B_4, B_6, D_4, D_6$ [7, Lemmas 3.5, 4.4].

We have not re-run these calculations. Each $\varphi \in \mathrm{Aut}(A)$ thus induces $\varphi_W \in \mathrm{Aut}(W)$ and $\theta_\varphi := \theta_{\varphi_W}$.

**Lemma 2.3.** *For any Coxeter group, $s \mapsto (e_s, s)$ defines a homomorphism $f\colon A \to \mathbb{Z}^{\mathcal{T}} \rtimes W$, with $(v,w)(v',w') = (v + w\cdot v', ww')$ and $w \cdot e_r = e_{wrw^{-1}}$.*

*Proof.* For $\Pi(s,t;m)$, the $\mathbb{Z}^{\mathcal{T}}$-component is $\sum_{k=1}^{m} e_{r_k}$ with $r_k = (st)^{k-1}s$. For $\Pi(t,s;m)$ it is $\sum_k e_{r'_k}$ with $r'_k = (st)^{m-k}s$. Both sets equal $\{(st)^j s : 0 \leq j < m\}$. Compare [7, Lemma 3.7]. □

**Proposition 2.4.** *Let $A$ be of spherical type. Then $f(P) = 2\mathbb{Z}^{\mathcal{T}}$, and $f$ induces an isomorphism $H_1(P;\mathbb{Z}) \cong 2\mathbb{Z}^{\mathcal{T}}$ under which conjugation by $a$ acts as the permutation $\rho(\bar{a})$.*

*Proof.* We have $f(as^2a^{-1}) = (2e_{\bar{a}s\bar{a}^{-1}}, 1)$, and these elements generate $f(P)$. Moreover $P \cong \pi_1$ of the complexified reflection-arrangement complement [2], whose $H_1$ is free on the meridians, one per hyperplane [3, 27]. So $H_1(P;\mathbb{Z})$ has rank $|\mathcal{T}|$, and the surjection onto $2\mathbb{Z}^{\mathcal{T}}$ is an isomorphism. □

**Corollary 2.5.** *Let $M_\varphi$ be the matrix of $\varphi_*$ on $\mathbb{Z}^{\mathcal{T}}$. Then $M_\varphi \rho(w) M_\varphi^{-1} = \theta_\varphi(\rho(w))$ and $M_{\iota_a\circ\varphi} = \rho(\bar{a})M_\varphi$. Hence $\theta_\varphi$ preserves the permutation character $\chi_{\mathcal{T}}(g) = \#\{r : g(r) = r\}$.*

The image $f(A) \cong A/[P,P]$ is the crystallographic quotient of [22, 26]; see also [13].

## 3. The fixed-vector criterion

A $\rho(W)$-invariant subspace $U \subseteq \mathbb{Q}^{\mathcal{T}}$ is *good* if $U \otimes \mathbb{C}$ is irreducible and occurs with multiplicity one in $\mathbb{C}^{\mathcal{T}}$. A good $U$ is a full isotypic component.

**Lemma 3.1.** *If $U$ is good and $N \in GL(\mathbb{Z}^{\mathcal{T}})$ commutes with $\rho(W)$, then $NU = U$ and $N|_U = \pm 1$.*

*Proof.* $NU \cong U$, so $NU = U$. By Schur's lemma $N|_U = c \in \mathbb{Q}^\times$. Applying $N^{\pm 1}$ to a primitive vector of $U \cap \mathbb{Z}^{\mathcal{T}}$ gives $c^{\pm 1} \in \mathbb{Z}$. □

**Lemma 3.2.** *A nonzero invariant $U \subseteq \mathbb{Q}^{\mathcal{T}}$ is good in each of the following cases.*

*(i) Every orbital matrix of $\rho(W)$ on $\mathcal{T} \times \mathcal{T}$ acts on $U$ by a scalar.*
*(ii) $\langle \chi_U, \chi_U \rangle = \langle \chi_U, \chi_{\mathcal{T}} \rangle = 1$.*
*(iii) $U \otimes \mathbb{C}$ is irreducible and $\sum_r \dim(U \otimes \mathbb{C})^{\mathrm{Stab}(r)} = 1$, where $r$ runs over representatives of the $\rho(W)$-orbits on $\mathcal{T}$ (Frobenius reciprocity).*

*Proof.* (i) The orbital matrices span the commutant $\mathcal{C} = \mathrm{End}_{\mathbb{C}W}(\mathbb{C}^{\mathcal{T}}) = \prod_i M_{m_i}(\mathbb{C})$, where $\mathbb{C}^{\mathcal{T}} = \bigoplus_i V_i \otimes \mathbb{C}^{m_i}$. By hypothesis $\mathcal{C}$ maps $U_{\mathbb{C}} = U \otimes \mathbb{C}$ into itself and acts on it by scalars. The central idempotents $e_i \in \mathcal{C}$ act on $U_{\mathbb{C}}$ by 0 or 1 and sum to 1, so exactly one $e_i$ acts as 1. Hence $U_{\mathbb{C}} \subseteq V_i \otimes \mathbb{C}^{m_i}$ meets only one isotypic component. Being $W$-invariant, $U_{\mathbb{C}} = V_i \otimes L$ for some $0 \neq L \subseteq \mathbb{C}^{m_i}$, and $M_{m_i}(\mathbb{C})$ preserves $L$ and acts on it by scalars. This forces $m_i = 1$ and $L = \mathbb{C}$. (ii) and (iii) are standard. □

**Theorem 3.3.** *Let $A$ be irreducible of spherical type. Suppose that for every $\varphi \in \mathrm{Aut}(A)$ there is a permutation $\pi$ of $\mathcal{T}$ with $\pi\rho(w)\pi^{-1} = \theta_\varphi(\rho(w))$ for all $w$, such that one of the following holds.*

*(i)* (stable) *There is a good $U$ with $\pi U = U$ such that, for each $\varepsilon = \pm 1$, some $h\pi$ ($h \in \rho(W)$) has eigenvalue $\varepsilon$ on $U$.*
*(ii)* (swap) *There are good $U_1 \neq U_2$ with $\pi U_1 = U_2$ such that, for each $\varepsilon = \pm 1$, some $(h\pi)^2$ has an eigenvector in $U_1$ with eigenvalue $\varepsilon$.*

*Then $A$ has property $R_\infty$.*

*Proof.* $N := \pi^{-1} M_\varphi$ commutes with $\rho(W)$.

*Case (i).* $N|_U = c = \pm 1$. Choose $h = \rho(\bar{a})$ and $0 \neq v \in U$ with $h\pi v = cv$, and let $\varphi' = \iota_a \circ \varphi$. Then $M_{\varphi'} v = h\pi N v = v$.

*Case (ii).* $N|_{U_i} = c_i$. Choose $h = \rho(\bar{a})$ and $0 \neq v \in U_1$ with $(h\pi)^2 v = c_1 c_2 v$. With $Q = h\pi$ we have $QU_1 = U_2$, and $M_{\varphi'}^2 v = QNQNv = v$. Since $M_{\varphi'} v \in U_2$ and $U_1 \cap U_2 = 0$, $u = v + M_{\varphi'} v$ is a nonzero fixed vector.

So $M_{\varphi'}$ fixes a nonzero vector of $\mathbb{Z}^{\mathcal{T}}$. By Lemma 2.1(b),(a) and Proposition 2.4, $R(\varphi'|_P) = \infty$. Then $R(\varphi') = \infty$ by (d), and $R(\varphi) = R(\varphi')$ by (c). □

**Remark 3.4** (Relation to the crystallographic criteria)**.** The image $f(A) \cong A/[P,P]$ is an extension of $W$ by the lattice $2\mathbb{Z}^{\mathcal{T}}$. If $Z(W) = 1$, it is a crystallographic group with holonomy representation $\rho$. In that case Theorem 3.3 is essentially the Dekimpe–Penninckx criterion [11, Cor. 3.10], [10, Thm. 3.3]: $R = \infty$ iff $\det(1 - AD) = 0$ for some holonomy element $A$. It is applied to $f(A)$, with the multiplicity-one device of [8, Thm. 4.7] used to produce the eigenvector. If $Z(W) \neq 1$, the holonomy action has kernel $Z(W)$, so $f(A)$ is not crystallographic in the strict sense. Working directly with $P$ avoids this issue.

**Remark 3.5.**

- If $\pi$ works for $\theta$, then $g\pi$ works for $\iota_g\theta$, since $\{hg\pi\} = \{h\pi\}$. So one $\theta$ per coset of $\mathrm{Inn}(\rho(W))$ suffices.
- For inner $\theta$ take $\pi = 1$. Then (i) holds for any good $U$ on which some reflection acts nontrivially. This is automatic for a nontrivial good $U$ when all reflections are conjugate: the kernel of the action is normal, so if it contained one reflection it would contain all of them.
- If $\theta(\rho(\mathcal{T})) = \rho(\mathcal{T})$, then $\pi(r) = \rho^{-1}\theta\rho(r)$ realises $\theta$.

## 4. The exceptional types

4.1. **Induced automorphisms.** Let $W$ be finite irreducible. Let $R(W) \leq \mathrm{Aut}(W)$ be the reflection-preserving automorphisms and $\mathrm{Gr}(W)$ the graph automorphisms. A *central twist* $w \mapsto w\,c(w)$, with $c\colon W \to Z(W)$ a homomorphism, induces the identity on $W/Z(W)$. Franzsen and Howlett prove the following ([21, Thm. 31 and the discussion preceding it, pp. 329–330]).

- For $W$ of type $B_n, D_n, E_6, E_7, E_8, F_4, H_3, H_4$, $\mathrm{Aut}(W) = R(W)\mathcal{A}(W)$, where $\mathcal{A}(W)$ consists of central twists.
- $R(W) = \mathrm{Inn}(W)\,\mathrm{Gr}(W)$ except for $H_3$ and $H_4$, where $[R(W) : \mathrm{Inn}(W)] = 2$.

In [21] (in the published version as in the preprint) the proof of Theorem 31 is given as an outline, based on the sizes of the conjugacy classes of involutions. (In item (9) of that theorem, for $F_4$, "$\alpha_\chi$" should read $\alpha_\zeta$.) For the six exceptional types, Appendix B gives independent control of all of $\mathrm{Aut}(\rho(W))$: by exhaustive computation for $E_6, E_7, F_4, H_3, H_4$, and by hand for $E_8$.

**Lemma 4.1.** *Let $\varphi \in \mathrm{Aut}(A)$, with $A$ of type $X$. Then $\theta_\varphi \in \mathrm{Inn}(\rho(W))\,\Theta_X$, where:*

- $\Theta_X = \{1\}$ *for* $X = E_6, E_7, E_8, B_n$ *($n \geq 3$);*
- $\Theta_{F_4} = \{1, \theta_L\}$, $\Theta_{H_3} = \{1, \theta_\Sigma\}$, $\Theta_{H_4} = \{1, \theta_\Psi\}$;
- $\Theta_{D_n} = \{\theta_\gamma : \gamma \in \mathrm{Gr}(D_n)\}$.

*Every element of $\Theta_X$ maps $\rho(\mathcal{T})$ onto itself, so it is realised by a permutation of $\mathcal{T}$.*

*Proof.* Central twists act trivially on $\rho(W)$, so $\theta_\varphi$ is induced by an element of $R(W)$.

- For $E_6$ (and $D_n$, $n$ odd) the graph automorphism is induced by conjugation by $w_0$, hence inner. For $E_7$, $E_8$ and $B_n$, Gr is trivial.
- For $F_4$, conjugation by the similarity $L(x) = (x_1 + x_2, x_1 - x_2, x_3 + x_4, x_3 - x_4)$ satisfies $Ls_\alpha L^{-1} = s_{L\alpha}$. It is reflection-preserving and exchanges the two reflection classes, so it is not inner. Since $[R(W) : \mathrm{Inn}(W)] \leq |\,\mathrm{Gr}(W)| = 2$, it represents the nontrivial coset.
- For $H_3$, resp. $H_4$, conjugation by $\Sigma$, resp. $\Psi$ (Propositions 4.3 and 4.6) is reflection-preserving and induces a non-inner automorphism of $\rho(W)$. It therefore represents the nontrivial coset of $\mathrm{Inn}(W)$ in $R(W)$.

Finally, $\theta_r(\rho(\mathcal{T})) = \rho(\mathcal{T})$ for $r \in R(W)$, by definition of $R(W)$. □

4.2. **Explicit good constituents.**

**Proposition 4.2** ($E_6, E_7, E_8$)**.** *Let $A_\perp$ be the adjacency matrix of the orthogonality graph on root lines.*

*(a) $\rho(W)$ has exactly three orbitals on $\mathcal{T} \times \mathcal{T}$: equality, orthogonality and non-orthogonality.*
*(b) The graph is strongly regular with parameters $(36, 15, 6, 6)$, $(63, 30, 13, 15)$, $(120, 63, 30, 36)$. Its nontrivial eigenvalues are $3^{15}, (-3)^{20}$; $3^{35}, (-5)^{27}$; $3^{84}, (-9)^{35}$ (multiplicities as exponents).*
*(c) The two nontrivial eigenspaces are good, and every reflection acts nontrivially on them.*

*Proof.* By Steinberg's theorem [33], the stabiliser of a root, or of a pair of roots, is the Weyl group of the perpendicular root subsystem. The perpendicular subsystems are listed below. The root counts and ranks were checked by machine (Appendix A). Each pair (number of roots, rank) in the table occurs for exactly one simply-laced root system, irreducible or not, of that rank, so the counts determine the types. These are the classical facts that the centraliser of a reflection in $W(E_8)$, $W(E_7)$, $W(E_6)$ involves $W(E_7)$, $W(D_6)$, $W(A_5)$.

| | $\alpha^\perp$ | $\{\alpha, \gamma\}^\perp$, $\alpha \perp \gamma$ | $\{\alpha, \beta\}^\perp$, $(\alpha, \beta) = 1$ |
|---|---|---|---|
| $E_6$ | $A_5$ (30, rank 5) | $A_3$ (12, rank 3) | $A_2^2$ (12, rank 4) |
| $E_7$ | $D_6$ (60, rank 6) | $D_4A_1$ (26, rank 5) | $A_5$ (30, rank 5) |
| $E_8$ | $E_7$ (126, rank 7) | $D_6$ (60, rank 6) | $E_6$ (72, rank 6) |

$W$ is transitive on roots. Inside $W(\alpha^\perp)$, the orbits are as follows.

- The orbit of an orthogonal root has size $720/24 = 30$, $23040/384 = 60$, $2903040/23040 = 126$, i.e. all of $\alpha^\perp$.
- The orbit of a root $\beta$ with $(\alpha, \beta) = 1$ has size $720/36 = 20$, $23040/720 = 32$, $2903040/51840 = 56$, i.e. all such roots.

Passing to lines gives (a). Then $k$, $\lambda$, $\mu$ are half the root counts in the table, and (b) follows from the standard formulas for strongly regular graphs.

By (a) the commutant is spanned by $I$, $A_\perp$ and $J$. These act by scalars on the nontrivial eigenspaces, which are rational because the eigenvalues are integers. So Lemma 3.2(i) applies, and Remark 3.5 gives the last assertion. □

**Proposition 4.3** ($H_3$)**.** *(a) $\rho(W) \cong A_5$. The 15 root lines form 5 orthogonal triples (*frames*), which correspond to the Klein four-subgroups.*

*(b) The frame module $U$ (functions constant on frames, with sum 0) is a good 4-dimensional constituent. More precisely, writing irreducible $A_5$-modules by their dimensions,*

$$\mathbb{Q}^{\mathcal{T}} \cong 1 \oplus U \oplus 5 \oplus 5.$$

*(c) For $\Sigma = \varsigma \circ \sigma$, where $\sigma$ is Galois conjugation $\sqrt{5} \mapsto -\sqrt{5}$ on coordinates and $\varsigma$ swaps the last two coordinates, $\Sigma\Phi = \Phi$ and $\Sigma s_\alpha \Sigma^{-1} = s_{\Sigma\alpha}$. The induced $\pi_\Sigma$ is not inner on $\rho(W)$.*

*(d) For every outer $\theta$ there is $h$ such that $h\pi$ acts on the frames as a transposition. It then has eigenvalues $+1$ (multiplicity 3) and $-1$ (multiplicity 1) on $U$.*

*Proof.* (a) Commuting reflections correspond to orthogonal lines, and the 15 involutions of $A_5$ lie in 5 Klein four-subgroups, permuted as $A_5$ permutes 5 points.

(b) $U$ is the augmentation module of a 2-transitive action, hence absolutely irreducible. Since $\mathbb{Q}^{\mathcal{T}} = \mathrm{Ind}_{V_4}^{A_5} 1$, the multiplicity of an irreducible $\chi$ is $\dim \chi^{V_4}$. This equals $1, 0, 0, 1, 2$ for $\chi$ of degree $1, 3, 3', 4, 5$, which gives the displayed decomposition.

(c) $\Sigma$ satisfies $(\Sigma x, \Sigma y) = \sigma(x, y)$. $\theta_\Sigma$ is not inner: on the rotation subgroup $W^+ \cong \rho(W)$ it Galois-conjugates the character of $\mathbb{R}^3$ on elements of order 5, while inner automorphisms preserve characters. This is also checked by machine.

(d) An outer automorphism of $A_5$ is conjugation by an odd permutation, and $\pi$ permutes the frames accordingly. So $\{h\pi\}$ contains a transposition. In fact $\pi_\Sigma$ itself is a transposition on frames. □

**Proposition 4.4** ($F_4$)**.** *Each class of 12 root lines (long, short) splits into 3 orthogonal frames of 4 lines. Let $U_{\mathrm{long}}, U_{\mathrm{short}}$ be the corresponding frame modules (sum zero). Then:*

*(a) both are good and 2-dimensional;*

*(b) long reflections act on $U_{\mathrm{long}}$ as transpositions of frames and trivially on $U_{\mathrm{short}}$, and vice versa;*

*(c) $\pi_L$ exchanges $U_{\mathrm{long}}$ and $U_{\mathrm{short}}$, and $\pi_L^2 = 1$;*

*(d) for a long simple reflection $s$, $(s\pi_L)^2 = s \cdot \pi_L s \pi_L$ has eigenvalue $-1$ on $U_{\mathrm{short}}$, and $\pi_L^2$ gives eigenvalue $+1$.*

*Hence case (ii) of Theorem 3.3 holds for $\theta_L$.*

*Proof.* The long roots $\pm e_i \pm e_j$ form a $D_4$, whose frames correspond to the pairings $ij|kl$. The short lines $e_i$ form one frame. The lines $\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1)$ form two frames, according to the parity of the number of minus signs. Each frame system is permuted through all of $S_3$, so each frame module is the standard module of $S_3$.

*Multiplicity one* (Lemma 3.2(iii)):

- the stabiliser of the long line $e_1 + e_2$ fixes one long frame and contains $s_{e_1 - e_2}$, which swaps the other two, so $\dim U_{\mathrm{long}}^{\mathrm{Stab}} = 1$;

- the stabiliser of the short line $e_1$ contains $\mathrm{Sym}\{2,3,4\}$, which induces $S_3$ on pairings, so $\dim U_{\mathrm{long}}^{\mathrm{Stab}} = 0$;
- $U_{\mathrm{short}}$ follows by applying $\pi_L$.

(b) Check on $s_{e_1-e_2}$ and $s_{e_1}$, and use conjugacy.
(c) $L$ maps orthogonal frames to orthogonal frames and long to short lines, and $L^2 = 2I$.
(d) $\pi_L s \pi_L$ is a short reflection. □

**Lemma 4.5** (Icosian model of $H_4$)**.** *Halve the* 120 *roots of* $H_4$ *(Appendix A) and read the first coordinate as the real part. These form a group* $I$ *of unit quaternions (the binary icosahedral group), and* $K = I/\{\pm 1\} \cong A_5$*. Identify* $\mathcal{T}$ *with* $K$*. Then:*

*(a) the reflection in a unit root* $r$ *is* $x \mapsto -r\bar{x}r$*, and it acts on* $K$ *by* $x \mapsto rx^{-1}r$*;*
*(b)* $\rho(W) = \{x \mapsto ax^{\pm 1}b : a, b \in K\} \cong (K \times K) \rtimes C_2$*.*

*Proof.* (a) For $|r| = 1$, $r\bar{x}r = 2(x,r)r - x$, using $\bar{x}r + \bar{r}x = 2(x,r)$.
(b) By (a), $\rho(W)$ is contained in the right-hand group, which has order $7200 = |W|/2 = |\rho(W)|$. The closure of $I$ and the formulas are also checked by machine. □

**Proposition 4.6** ($H_4$)**.** *Let* $V$ *be the* 4*-dimensional augmentation module of* $K \cong A_5$ *acting on its five Klein four-subgroups, and let* $U \subseteq \mathbb{Q}^K$ *be the span of the matrix coefficients* $x \mapsto \mathrm{tr}(X\lambda_V(x))$*, so* $U \cong \mathrm{End}(V)$ *has dimension* 16*.*

*(a)* $U$ *is good, and* $\mathbb{Q}^{\mathcal{T}} \cong 1 + (3 \otimes 3) + (3' \otimes 3') + 16 + 25$*.*
*(b) Let* $\Psi = \Omega \circ \sigma$*, where* $\sigma$ *is Galois conjugation and* $\Omega$ *is the rotation* $j \mapsto k$*,* $k \mapsto -j$*. Then* $\Psi$ *is a ring automorphism of* $\mathbb{H}_{\mathbb{Q}(\sqrt{5})}$ *with* $\Psi(I) = I$*. The induced permutation* $\pi_\Psi$ *normalises* $\rho(W)$*, preserves reflections, and is not inner.*
*(c) For every outer* $\theta$ *there is* $h$ *such that* $h\pi$ *acts on* $U$ *as conjugation by a transposition. This has eigenvalues* $+1$ *(multiplicity* 10*) and* $-1$ *(multiplicity* 6*).*

*Proof.* (a) By Wedderburn, $\mathbb{C}[K] = \bigoplus_{V'} V' \otimes V'^*$ as a $K \times K$-module, each summand irreducible and occurring once. The summand for $V$ is $U$, which is rational and stable under inversion. The decomposition follows from the irreducible degrees $1, 3, 3, 4, 5$ of $A_5$.

(b) $\Psi$ commutes with quaternion conjugation, so $\Psi s_r \Psi^{-1} = s_{\Psi r}$. Closure $\Psi(I) = I$ is checked exactly. If $\theta_\Psi$ were inner, then $\pi_\Psi \in \rho(W)$, because the centraliser of $\rho(W)$ in $\mathrm{Sym}(K)$ is trivial. Since $\pi_\Psi$ fixes 1, it would be $x \mapsto ax^{\pm 1}a^{-1}$. The sign $-$ is impossible, since $\pi_\Psi$ is an automorphism of the non-abelian group $K$. The sign $+$ is impossible, since conjugation preserves real parts up to sign, whereas $\Psi$ maps an element of order 5 with real part $\tau/2$ to one with real part $-\tau^{-1}/2$ (here $\tau = \frac{1+\sqrt{5}}{2}$).

(c) We argue in three steps.

(i) By Lemma 4.1 and Remark 3.5, for an outer $\theta$ we may take $\pi \in \rho(W)\pi_\Psi$. So $\{h\pi : h \in \rho(W)\} = \{h\pi_\Psi\}$.
(ii) By (b), $\pi_\Psi = \psi$ is an outer automorphism of $K$.
(iii) Since $\mathrm{Aut}(A_5) \cong S_5$, acting on the five Klein four-subgroups, $\psi$ is conjugation by an odd permutation $\delta'$. For $a \in K = A_5$, the element $h = (x \mapsto axa^{-1})$ lies in $\rho(W)$, and $h\psi$ is conjugation by $a\delta'$. Choose $a$ with $a\delta' = \delta$ a transposition.

The action of $K$ on $V$ extends to $S_5$, with $\lambda_V(\delta)$ having eigenvalues $1, 1, 1, -1$. Since $h\pi_\Psi$ acts on $U \cong \mathrm{End}(V)$ as $X \mapsto \lambda_V(\delta)X\lambda_V(\delta)^{-1}$, the eigenvalue $+1$ has multiplicity $9 + 1$ and $-1$ has multiplicity 6. □

*Proof of the Main Theorem.* By Lemma 4.1 and Remark 3.5, it suffices to treat $\Theta_X$.

- $E_n$: only the inner class occurs. Proposition 4.2 supplies $U$, and Remark 3.5 applies, since all reflections are conjugate.

- $H_3$, $H_4$: the inner class is covered by Remark 3.5 (all reflections are conjugate and $U$ is nontrivial). The outer class is covered by Propositions 4.3 and 4.6: there $\pi U = U$, since $U$ is the only good constituent of its dimension, and case (i) applies.
- $F_4$: there are two classes of reflections, so the inner class is treated explicitly. Take $\pi = 1$ and $U = U_{\mathrm{long}}$; then $h = 1$ gives $+1$, and a long reflection gives $-1$ by Proposition 4.4(b), so case (i) holds. The outer class $\theta_L$ is covered by Proposition 4.4, case (ii).
- $D_5$: Proposition 5.2.

Apply Theorem 3.3. □

| type | $\lvert\mathcal{T}\rvert$ | $\lvert\rho(W)\rvert$ | $\Theta_X$ | good constituent used (dim) | outer witness |
|---|---|---|---|---|---|
| $E_6$ | 36 | 51840 | 1 | **SRG** eigenspace (15; also 20) | — |
| $E_7$ | 63 | 1451520 | 1 | **SRG** eigenspace (35; also 27) | — |
| $E_8$ | 120 | 348364800 | 1 | **SRG** eigenspace (84; also 35) | — |
| $F_4$ | 24 | 576 | $\{1, \theta_L\}$ | $\mathbf{U_{long}}, \mathbf{U_{short}}$ (2+2) | swap: $\pi_L^2$, $(s\pi_L)^2$ |
| $H_3$ | 15 | 60 | $\{1, \theta_\Sigma\}$ | **frame** module (4) | eigen-multiplicities 3, 1 |
| $H_4$ | 60 | 7200 | $\{1, \theta_\Psi\}$ | End(**V**) (16) | eigen-multiplicities 10, 6 |

TABLE 1. The data used for the Main Theorem. The dimensions agree with independent computer certificates (Appendix A).

## 5. Uniform proofs for the classical types

Except for $D_5$, these results are known [5, 30]. We include them to illustrate the method, and because for $D_5$ the argument gives what appears to be the first proof (Remark 5.3).

**Proposition 5.1** ($A_n$, $n \geq 2$)**.** *$A(A_n)$ has property $R_\infty$.*

*Proof.* Here $\rho(W) = S_{n+1}$ acts on the transpositions, i.e. on the 2-subsets. By Young's rule $\mathbb{Q}^{\mathcal{T}} = S^{(n+1)} \oplus S^{(n,1)} \oplus S^{(n-1,2)}$ (for $n = 2$ the last summand is absent). So $S^{(n,1)}$ is good. For $n+1 \neq 6$, $\mathrm{Aut}(S_{n+1}) = \mathrm{Inn}$. For $n+1 = 6$, an outer automorphism maps $(12) \mapsto (12)(34)(56)$, but $\chi_{\mathcal{T}}$ takes the values 7 and 3 there, so it is not of the form $\theta_\varphi$ (Corollary 2.5). Apply Remark 3.5. □

For $B_n$ and $D_n$ let $u_{ij} = e_{[e_i+e_j]} - e_{[e_i-e_j]}$ $(i < j)$ and $U^- = \mathrm{span}\{u_{ij}\}$. Let $E \leq W$ be the group of sign changes: all of them for $B_n$, the even ones for $D_n$.

**Proposition 5.2** ($B_n$, $n \geq 3$; $D_n$, $n \geq 5$)**.** *$U^-$ is good, and $A(B_n)$ and $A(D_n)$ have property $R_\infty$.*

*Proof. Weights.* $E$ acts on $u_{ij}$ by $\chi_{ij}(\varepsilon) = \varepsilon_i\varepsilon_j$. These characters are nontrivial and pairwise distinct on $E$. For $D_n$ this needs $n \geq 5$: on even sign changes $\chi_{ij} = \chi_{kl}$ iff $\varepsilon_i\varepsilon_j\varepsilon_k\varepsilon_l$ is trivial or the product of all signs.

*Goodness.* $E$ fixes $e_{[e_i+e_j]} + e_{[e_i-e_j]}$ and the short lines. So each $\chi_{ij}$ occurs once in $\mathbb{Q}^{\mathcal{T}}$. Since $S_n \leq W$ permutes the weights transitively, $U^- \otimes \mathbb{C}$ is irreducible.

*Automorphisms.* By Lemma 4.1, $\theta_\varphi$ is inner for $B_n$, and inner up to the diagram involution for $D_n$ (for odd $n$ the latter is inner too). For the inner class, $\pi = 1$ works: the reflection $s_{e_n}$ $(B_n)$, resp. $s_{e_{n-1}-e_n}$ $(D_n)$, acts nontrivially on $U^-$. The diagram involution is realised by negating $e_n$, which acts on $u_{ij}$ by $-1$ if $n \in \{i, j\}$ and by $+1$ otherwise. So case (i) holds with $h = 1$. □

**Remark 5.3** (Type $D_5$)**.** Proposition 5.2 with $n = 5$ proves $R_\infty$ for $A(D_5)$. Its inputs are [7], [21] (here $\mathrm{Aut}(W(D_5)) = R(W) \cong W$ by [21, Thm. 31(3)]) and the weight argument.

There is also a short derivation from Paris–Soroko's description $\mathrm{Aut}(A[D_5]) = \mathrm{Inn}(A) \rtimes \langle\chi\rangle$, with $\chi$ the inversion of the generators [28, Thm. 1]:

- $R(\iota_g) = R(\mathrm{id}) = \infty$ by Lemma 2.1(c), since $A^{ab} \cong \mathbb{Z}$ and Lemma 2.1(a),(b) give infinitely many conjugacy classes;
- $R(\iota_g \circ \chi) = R(\chi) = \infty$ by Remark 5.6(b).

This deduction is nearly immediate once [28, Thm. 1] is available, but we have not found it written down. [30] states $D_5$ as a conjecture, and [28] does not mention $R_\infty$. We therefore record $D_5$ as part of the Main Theorem.

**Proposition 5.4** ($D_4$)**.** *In type $D_4$:*

- *$U^-$ is reducible: $J(u_{ij}) = u_{kl}$ ($\{k,l\}$ complementary) commutes with $W(D_4)$, and $U^-$ is the sum of two good 3-dimensional constituents;*
- *the frame module $U$ on the 3 orthogonal frames $\{12|34, 13|24, 14|23\}$ is good of dimension 2;*
- *$A(D_4)$ has property $R_\infty$.*

*Proof.* The $E$-fixed part of $\mathbb{Q}^{\mathcal{T}}$ is the permutation module on the 6 pairs, namely $1+3+2$ for $S_4$. Its 2-dimensional summand is $U$, which does not occur in $U^-$. Every $\theta_\varphi$ preserves reflections, so $\pi$ permutes the frames. $W$ induces all of $S_3$ on frames, so some $h\pi$ is a transposition there, with eigenvalues $\pm 1$ on $U$. □

**Remark 5.5** (Dihedral types)**.** Let $m$ be odd. Then $Z(W) = 1$, so $\rho$ is faithful and injective on $\mathcal{T}$, and $\mathbb{Q}^{\mathcal{T}} \cong \bigoplus_{d|m} U_d$ with $\dim U_d = |(\mathbb{Z}/d\mathbb{Z})^\times|$, and $U_d$ is absolutely irreducible iff $d \in \{1, 3\}$.

- If $3 \nmid m$ there is no nontrivial good constituent, and Theorem 3.3 cannot be applied. For $m$ prime, Schur's lemma gives only a unit of $\mathbb{Q}(\cos\frac{2\pi}{m})$.
- If $3 \mid m$, then $U_3$ is the unique good constituent of dimension 2, so $\pi U_3 = U_3$. The group $\rho(W)$ acts on $U_3$ as $S_3 \leq O(2)$. Since $\pi$ is orthogonal and normalises $S_3$, it acts as an element of $S_3 \times \{\pm 1\}$. Hence some $h\pi$ acts as $\pm$(a reflection), with eigenvalues $\pm 1$. So Theorem 3.3 applies, using [7] and that every automorphism of $W$ preserves reflections.

This was confirmed by computation for odd $m \leq 21$. We have not analysed even $m$. In any case $R_\infty$ for $I_2(m)$ is known [5].

**Remark 5.6** (Two elementary observations)**.** (a) If $\varphi$ fixes the generator $z \in \{\Delta, \Delta^2\}$ of $Z(A)$ [4, 12], then $\varphi_*$ fixes the nonzero image of $z$ in $H_1(A)$, so $R(\varphi) = \infty$. Only centre-inverting automorphisms need the criterion.

(b) For any Artin group that is not free abelian, the inversion $s \mapsto s^{-1}$ has $R = \infty$. Choose $m_{st} \geq 3$. Via $f$, $\max_r |v_r - v_{srs}|$ is an invariant of twisted classes with $W$-part $s$, and it separates the elements $t^{2k}s$. Cf. [29, Thm. 1], [24].

## Appendix A. Computational cross-checks

The following were checked by exact arithmetic (over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$). Programs, commands and expected outputs are listed in `code/README.md` of the accompanying repository.

*Coordinates.* For $E_8$ we use Bourbaki's simple roots in $\mathbb{R}^8$ with basis $e_1, \dots, e_8$:

$$\alpha_1 = \tfrac{1}{2}(e_1 + e_8) - \tfrac{1}{2}(e_2 + \cdots + e_7), \qquad \alpha_2 = e_1 + e_2, \qquad \alpha_k = e_{k-1} - e_{k-2} \quad (3 \leq k \leq 8).$$

$E_7$ and $E_6$ use $\alpha_1, \dots, \alpha_7$, resp. $\alpha_1, \dots, \alpha_6$, in $\mathbb{R}^8$. The other types:

- $F_4$: $e_2 - e_3$, $e_3 - e_4$, $e_4$, $\frac{1}{2}(e_1 - e_2 - e_3 - e_4)$;
- $A_n, B_n, D_n$: standard simple roots;
- $H_3$: roots $\pm 2e_i$ and the even permutations of $(\pm\tau, \pm 1, \pm\tau^{-1})$, with $\tau = \frac{1+\sqrt{5}}{2}$;
- $H_4$: roots $\pm 2e_i$, $(\pm 1, \pm 1, \pm 1, \pm 1)$ and the even permutations of $(\pm\tau, \pm 1, \pm\tau^{-1}, 0)$.

*Checks.*

(1) Proposition 4.2: SRG parameters, three orbitals, eigenspace dimensions, perpendicular root counts and ranks.

(2) Propositions 4.3–4.6 and Lemma 4.5:
- frames, and character norms $\langle\chi,\chi\rangle = \langle\chi,\chi_{\mathcal{T}}\rangle = 1$;
- the images of $W$ on frames ($A_5$, $S_3$);
- the action of $\pi_\Sigma$, $\pi_L$, $\pi_\Psi$, with eigenspace dimensions $(3,1)$, $(1,1)$, $(10,6)$;
- the closure of $I$, the reflection formula, and $\rho(W) = \{x \mapsto ax^{\pm1}b\}$.

(3) Propositions 5.2 and 5.4:
- the weight argument for $B_3$–$B_6$ and $D_4$–$D_7$, including its failure for $D_4$;
- the operator $J$ and the frame module of $D_4$;
- witnesses for all six graph automorphisms of $D_4$.

(4) Earlier, independent certificates for the six exceptional types:
- commutant certificates;
- character certificates written as fresh code;
- witnesses for abstractly computed outer automorphisms.

They give the same dimensions $15, 20$; $27, 35$; $35, 84$; $4$; $2+2$; $16, 25$.

(5) The criterion (for automorphisms preserving $\chi_{\mathcal{T}}$) checked uniformly for $A_2$–$A_7$, $B_3$–$B_7$, $D_4$–$D_7$, odd $I_2(m)$ with $m \le 21$, and the exceptional types other than $E_8$.

## Appendix B. Corroboration: all automorphisms of $W/Z(W)$

The argument above uses only automorphisms of $\rho(W)$ induced from $\mathrm{Aut}(W)$. As independent corroboration of [21, Thm. 31] for the exceptional types (whose proof there is an outline), we record the stronger statement about all of $\mathrm{Aut}(\rho(W))$:

- $\mathrm{Out}(\rho(W)) = 1$ for $E_6$, $E_7$, $E_8$;
- $|\mathrm{Out}(\rho(W))| = 2$ for $H_3$, $H_4$, $F_4$.

For $E_6, E_7, H_3, H_4, F_4$ this was verified by exhaustive computation in two ways: brute force over generator images, and involution classes followed by orbits of Coxeter tuples. For $E_8$ the following argument by hand gives it.

**Lemma B.1.** *Let $\Phi$ be an irreducible simply-laced root system of rank $\ge 3$ (so its diagram is a tree). Then every $\theta \in \mathrm{Aut}(\rho(W))$ with $\theta\rho(\mathcal{T}) = \rho(\mathcal{T})$ is inner up to a graph automorphism.*

*Proof.* Write $\theta\rho(s_i) = \rho(s_{\beta_i})$; this uses that $\rho$ is injective on $\mathcal{T}$. Orders of products agree in $W$ and in $\rho(W)$. Indeed, for roots $\beta \ne \pm\gamma$ the element $s_\beta s_\gamma$ is a plane rotation fixing a complement of dimension $\ge 1$, so no nontrivial power of it lies in $Z(W) \subseteq \{\pm1\}$; in particular $s_i s_j \ne -1$. Hence $(\beta_i,\beta_j) = 0$ or $\pm1$ according to the diagram. Choose the signs along the tree. Then $g\colon \alpha_i \mapsto \beta_i$ is an isometry, and $g\Phi = (gWg^{-1})\{\beta_i\} \subseteq \Phi$. So $g\{\alpha_i\}$ is a simple system. As $W$ is simply transitive on simple systems, $g = w\gamma$ with $w \in W$ and $\gamma$ an isometry permuting $\{\alpha_i\}$, i.e. $g \in W \rtimes \mathrm{Gr} = \mathrm{Aut}(\Phi)$ [32]. Compare [21, Thm. 18, Cor. 19]. □

**Proposition B.2.** *Every automorphism of $W(E_8)/\{\pm1\}$ is inner.*

*Proof.* $G^+ = W^+/\{\pm1\}$ is the commutator subgroup, of index 2. An involution outside $G^+$ lifts to $w$ with $w^2 = 1$ and $\det w = -1$; if $w^2 = -1$ then $\det w = 1$. By [6, Lemma 5] (provable via Steinberg's theorem), $w$ is a product of $k = \dim\ker(w+1)$ reflections in mutually orthogonal roots, and $k$ is odd. Replacing $w$ by $-w$, we may take $k \in \{1,3\}$. For $k = 3$, the $(-1)$-eigenspace of $w$ is spanned by the three roots, and contains only the 6 roots $\pm\beta_i$: a root $\sum c_i\beta_i$ has $2c_i = (\cdot,\beta_i) \in \{0,\pm1,\pm2\}$ and $\sum c_i^2 = 1$. So $w$ determines the triple, and $-w$ (with $k = 5$) is not of this form.

The products for $k = 3$ form one class of size $240 \cdot 126 \cdot 60/48 = 37800$, by transitivity along $E_8 \supset E_7 \supset D_6$. So the reflections form the unique class of size 120 outside $G^+$. Hence every automorphism preserves $\rho(\mathcal{T})$, and Lemma B.1 applies. □

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